Estimating Differences and Checking Your Answer
Answer Key

Grades 2–3   Number and Operations in Base Ten

Quick Answers

1. 200	4. 400	7. 600	10. 300
2. 130	5. 400, 409	8. 300, 316	
3. 340	6. 727, 448	9. 545	

Curriculum

Level: Grades 2–3   Number and Operations in Base Ten
3.NBT.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 13 tagged checks.

Common wrong answers

7. *If they got 500: rounded 349 up to 400 by looking at the ones digit instead of the tens digit*
9. *If they got 655: took the smaller digit from the larger one in every column instead of regrouping*

1. Library books: estimate $412 - 187$ to the nearest hundred.
412 rounds to 400 (the tens digit is 1, less than 5, so round down). 187 rounds to 200 (the tens digit is 8, which is 5 or more, so round up). Subtract the rounded numbers: $400 - 200 = 200$. About 200 books are still on the shelf. $\boxed{200}$
Grader's note: the exact difference is 225 books, so an estimate of 200 books is a good one. Accept the estimate only when *both* numbers were rounded — rounding just one is the most common error here.

2. Ferry seats: estimate $265 - 138$ to the nearest ten.
265 rounds to 270 (the ones digit is 5, so round up). 138 rounds to 140 (the ones digit is 8). Subtract the rounded numbers: $270 - 140 = 130$. The ferry could have carried about 130 more passengers. $\boxed{130}$
Grader's note: the exact difference is 127 passengers. Rounding to the nearest ten lands within 3 of the exact answer, much closer than rounding to the nearest hundred would.

3. Apples: check 175 apples left by adding back the 165 apples sold.
Addition undoes subtraction, so the answer plus the number taken away must give back the starting number. Add $175 + 165$: ones, $5 + 5 = 10$, write 0 and carry 1; tens, $7 + 6 + 1 = 14$, write 4 and carry 1; hundreds, $1 + 1 + 1 = 3$. $\boxed{340}$
Expected sentence: "The crate started with 340 apples."

4. Stadium seats: estimate $738 - 264$ to the nearest hundred.
738 rounds to 700 (tens digit 3). 264 rounds to 300 (tens digit 6). Subtract the rounded numbers: $700 - 300 = 400$. About 400 seats are filled. $\boxed{400}$
Grader's note: the exact difference is 474 seats. The estimate is about 74 low because both numbers rounded in the same direction that widens the gap — worth saying aloud, but 400 is still the correct estimate.

5. Recycling cans: estimate, then compute exactly.
(a) Estimate. 906 rounds to 900 and 497 rounds to 500, so $900 - 500 = 400$. The club needs about 400 more cans. $\boxed{400}$
(b) Exact. $906 - 497$: the ones need a group but the tens digit is 0, so trade a hundred first — 906 is 8 hundreds, 10 tens, 6 ones, then 8 hundreds, 9 tens, 16 ones. Now $16 - 7 = 9$; $9 - 9 = 0$; $8 - 4 = 4$. $\boxed{409}$
(c) Reasonable? Yes — 409 cans is only 9 away from the estimate of 400 cans, so the exact answer is reasonable.

6. Book pages: Ben's check.
(a) Check by adding. $552 + 175$: ones, $2 + 5 = 7$; tens, $5 + 7 = 12$, write 2 and carry 1; hundreds, $5 + 1 + 1 = 7$. $\boxed{727}$
The check gives 727 pages, *not* the 623 pages the book has, so Ben's answer is wrong.
(b) Exact. $623 - 175$: the ones need a group, so trade a ten — 6 hundreds, 1 ten, 13 ones. The tens now need a group too, so trade a hundred — 5 hundreds, 11 tens, 13 ones. Then $13 - 5 = 8$; $11 - 7 = 4$; $5 - 1 = 4$. $\boxed{448}$
Check the corrected answer: $448 + 175 = 623$ pages ✓

7. Farm eggs: estimate $851 - 349$ to the nearest hundred.
851 rounds to 900 (tens digit 5, so round up). 349 rounds to 300 — look at the *tens* digit, which is 4, so round down. Subtract the rounded numbers: $900 - 300 = 600$. About 600 eggs are still at the farm. $\boxed{600}$
Grader's note: the exact difference is 502 eggs. A student who wrote 500 rounded 349 up to 400 by looking at the ones digit 9 — send them back to the tens digit rule.

8. Theater tickets: estimate, then compute exactly.
(a) Estimate. 604 rounds to 600 (tens digit 0) and 288 rounds to 300 (tens digit 8), so $600 - 300 = 300$. About 300 adult tickets. $\boxed{300}$
(b) Exact. $604 - 288$: the ones need a group and the tens digit is 0, so trade a hundred — 5 hundreds, 10 tens, 4 ones — then trade a ten — 5 hundreds, 9 tens, 14 ones. Now $14 - 8 = 6$; $9 - 8 = 1$; $5 - 2 = 3$. $\boxed{316}$
316 adult tickets sits close to the estimate of 300, so the answer is reasonable. Check: $316 + 288 = 604$ tickets ✓

9. Rico's stickers: why 655 cannot be right, then the exact answer.
Why it is unreasonable. Round to the nearest hundred: $703 \approx 700$ and $158 \approx 200$, so the answer should be near $700 - 200 = 500$ stickers. Rico's 655 stickers is more than 150 away from that, so it is far too big.
What went wrong. In each column Rico took the smaller digit from the larger one instead of regrouping.
Correct work. $703 - 158$: the ones need a group and the tens digit is 0, so trade a hundred — 6 hundreds, 10 tens, 3 ones — then trade a ten — 6 hundreds, 9 tens, 13 ones. Now $13 - 8 = 5$; $9 - 5 = 4$; $6 - 1 = 5$. $\boxed{545}$
The exact answer 545 stickers is close to the estimate of 500 stickers ✓

10. Food drive: estimate $900 - 348 - 275$ to the nearest hundred.
Round every number first: 900 is already a hundred; 348 rounds to 300 (tens digit 4); 275 rounds to 300 (tens digit 7). Now subtract in order: $900 - 300 = 600$, then $600 - 300 = 300$. About 300 more cans are needed. $\boxed{300}$
Grader's note: the exact answer is 277 cans, so the estimate of 300 cans is close. Accept either subtracting one at a time or adding the two weeks first ($300 + 300 = 600$ rounded) — both give 300.